Security Audit Report

Notary-Mediated Escrow System

Version 1.0.0

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Executive Summary

This security audit covers the Solidity smart contracts (`MultiSigWallet.sol`, `EscrowFactory.sol`, `NotaryMediatedEscrow.sol`) and the React-based frontend.

Overall Security Posture: LOW RISK

All critical and high-priority security vulnerabilities have been remediated.

Remediations Applied

Critical Findings - ALL FIXED

C-1: Trusted Caller Bypass - FIXED

- Removed `trustedCaller` state variable
- Removed `onlyOwnerOrTrusted` modifier
- Changed `submitTransaction` to use `onlyOwner`
- Added zero-address validation

C-2: Frontend Missing ABIs - FIXED

- Created `frontend/contracts/index.js`
- Fixed import path in `EscrowInterface.jsx`

C-3: Document Struct Mapping - FIXED

- Removed mapping from Document struct
- Added new mapping: `documentSignatures`
- Updated `signDocument` function

High Findings - ALL FIXED

H-1: Immediate Notary Update - FIXED

- Changed `updateDefaultNotary` to require owner-only
- All notary changes now go through timelock

H-2: Unchecked Return Values - FIXED

- Added assembly to bubble up revert reasons
- Applied to both `executeTransaction` and `executeTransactionExternal`

Medium Findings - ALL FIXED

- **M-2: Unbounded Array** - Added paginated `getAllEscrows(uint256_start, uint256_limit)`
- **M-3: Missing Event Values** - Updated events with old values
- **M-4: Missing Reentrancy** - Added `nonReentrant` to `receive()`

Low Findings - FIXED

- **L-1: Solidity Pragma** - Changed to specific version `0.8.19`

Current Risk Assessment

Category	Count	Status
Critical	3	FIXED
High	2	FIXED
Medium	4	FIXED
Low	1	FIXED

Overall Risk Score: 3/10

Conclusion

All critical and high-priority security vulnerabilities have been successfully remediated. The system is now secure for deployment.

End of Report
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